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Lecture Notes «Matemātiskā analīze I»

CALCULUS I

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LECTURE 8.

FUNCTION ANALYSIS II

Concavity
Local Extrema
8. Homework

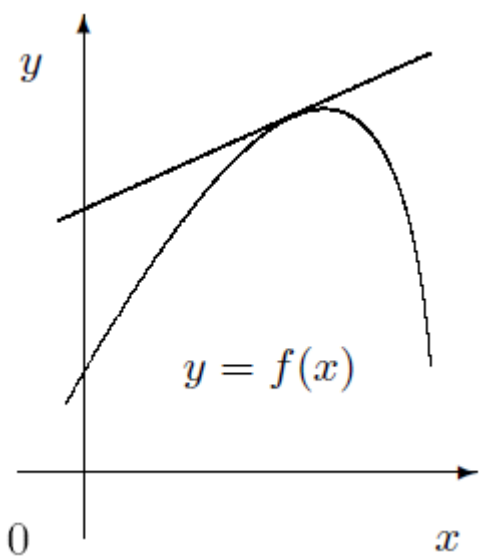
Concavity

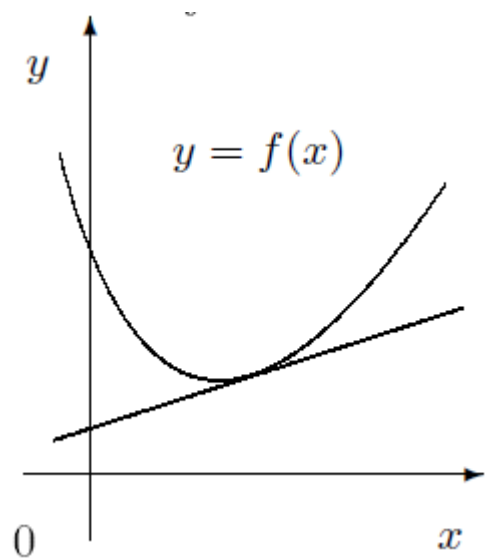
If the tangent line turns steadly in the counterclockwise direction, we say that the graph is concave up; if the tangent turns in the clockwise direction, the graph is concave down.

DEFINITION 8.1.

Let f be differentiable on an open interval I. We say that f (as well as its graph) is concave up (convex) on I if f' is increasing on I, and we say that f is concave down on I if f' is decreasing on I.

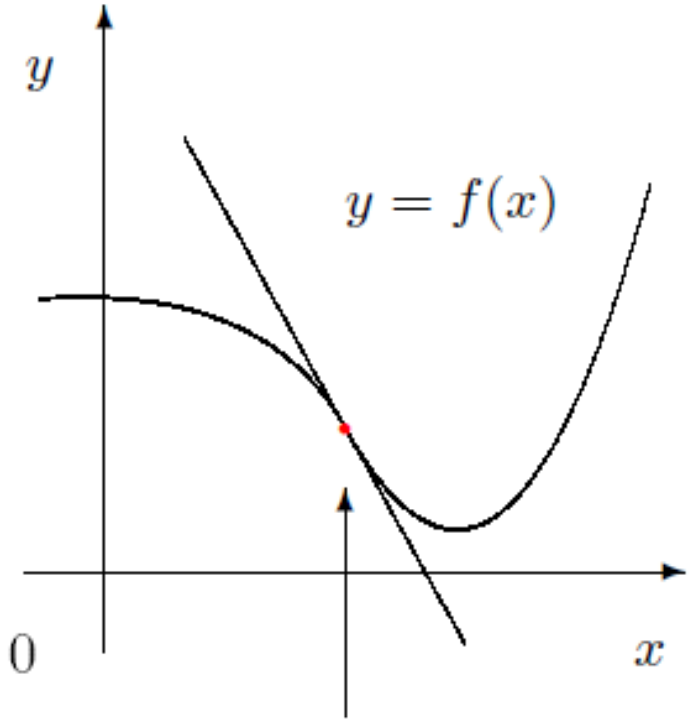
Concave up – the graph is **under** the tangent line in the considered interval
Concave down – the graph is **over** the tangent line in the considered interval





Concavity

DEFINITION 8.2.
Let f be continuous at c. We call $(c, f(c))$ an inflection point on the graph of f if f is concave up on one side of c and concave down on the other side.

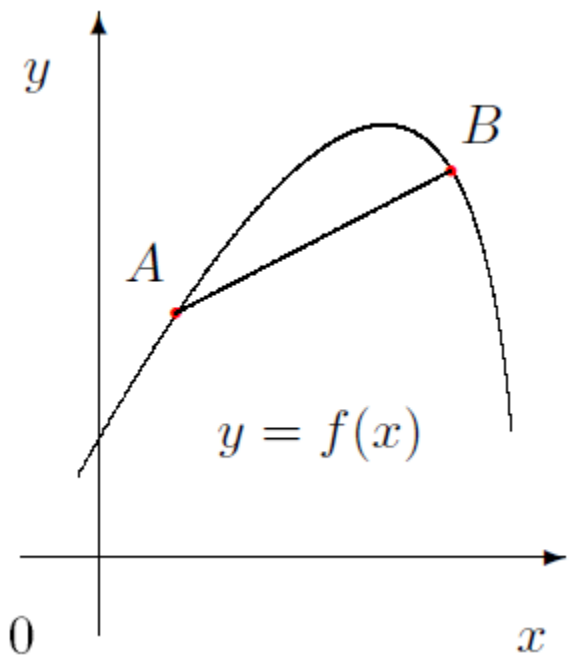


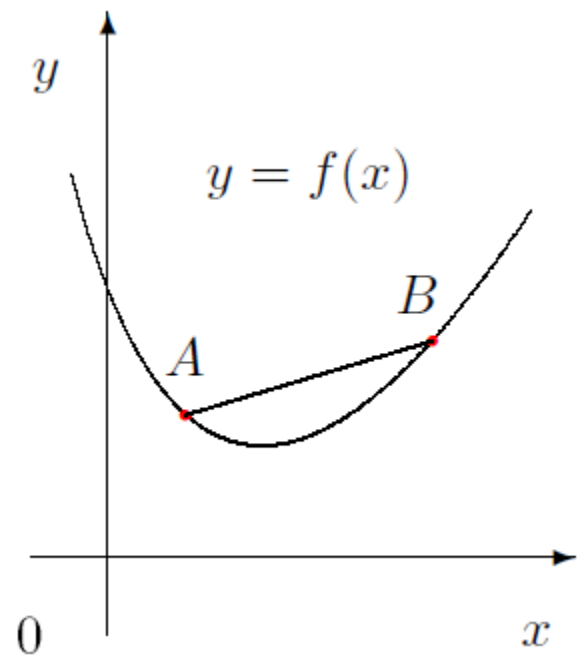
Concavity

Let *f* be differentiable on an open interval *I*

Concave up – for any two points of the interval the graph that connects these two points is **over** chord that connect these points

Concave down – for any two points of the interval the graph that connects these two points is **under** chord that connect these points





Concavity

THEOREM 8.3.
(Concavity theorem)
Let f be twice differentiable on an open interval I.
1) If $f''(x) < 0$ for all x in I, then f is concave up on I.
2) If $f''(x) > 0$ for all x in I, then f is concave down on I.

Points where $f''(x) = 0$ or where $f''(x)$ does not exist are the candidates for **points of inflection**. The conclusion is made according the change of signs of the f''(x).

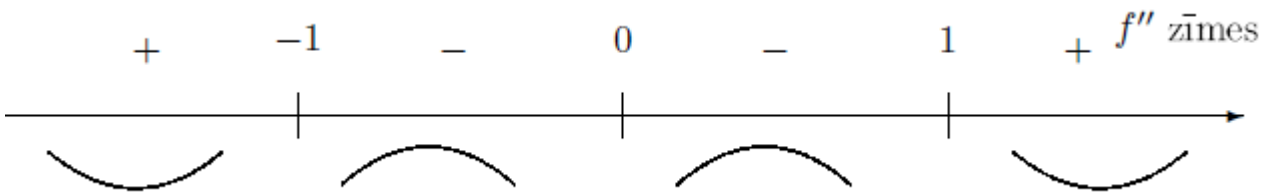
Concavity

Example

$$y = 3x^2 - \frac{1}{x^2}$$

$$y'' = 6\frac{x^4 - 1}{x^4}$$

Critical points are -1, 0 and 1.
Detect the signs of y''.



Function is
concave down on (-∞;-1) and (1; +∞)
concave up on (-1;0) and (0;1)

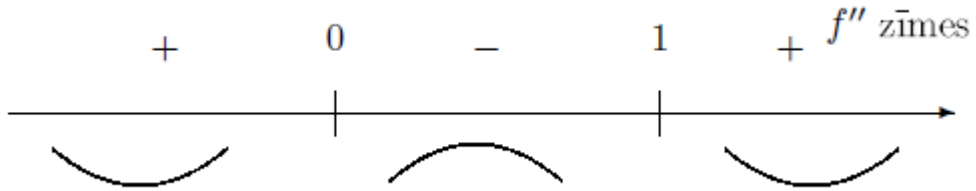
The **points of inflection** are (-1, 2) and (1, 2).
The point 0 does not belong to Dom(y) and also the change of signd does not indicate 0 as point of inflection.

Concavity

Example

$y = (x+2)\sqrt[3]{x}$
Please find the points of inflection!

$$y'' = \frac{4(x-1)}{9\sqrt[3]{x^5}}$$
Critical points are 0 and 1.
Detect the signs of y''.



Both critica points (0, 0) and (0, 3) are **points of inflection**.
The point 0 belongs to Dom(y), but at this point does not exist y''.

Function is
concave down on (-∞;-1) and (1; +∞)
concave up on (0;1)

Concavity

Exercises

Please find the points of inflection and concavity intervals!

$a)\ y = \dfrac{x^2}{(x-1)^2}$

$b)\ y = \ln(x^2 + 2x - 3)$

Local Extrema

The maximum and minimum value on a given interval are reffered as **global extreme values** (global maximum value and global minimum value).

DEFINITION 8.6. (Monotonicity theorem)
Let S, the domain of f, contain the point c. We say that

1) $f(c)$ is a local maximum value of f if there ia an interval (a, b) containing c such that $f(c)$ is the maximum value of f on $(a, b) \cap S$.
2) $f(c)$ is a local minimum value of f if there ia an interval (a, b) containing c such that $f(c)$ is the minimum value of f on $(a, b) \cap S$.
3) $f(c)$ is a local extreme value of f if it is either a local minimum or a local maximum value.

Any global extreme value is also a local extreme value.

Local Extrema

Where do local extreme values occur?

The candidates for points where local extreme may occure are critical points:

- **End** points
- **Stationary** points (the derivative = 0)
- **Singular** points (the derivative does not exist)

We are not claiming that there must be a local extremum at every critical point.

Example

$y = x^3, y' = 3x^2$, the only critical point is 0, but here the function does not reach local minimum or local maximum value.

However, if the derivative is positive on one side of the critical point and negative on the other (and if the function is continuous), then we have alocal extremum.

Local Extrema

THEOREM 8.7.

(First derivative test)

Let f be continuous on an open interval (a, b) that contains a critical point c.

1) If $f'(x) > 0$ for all x in (a, c) and $f'(x) < 0$ for all x in (c, b), then $f(c)$ is a local maximum value of f.
2) If $f'(x) < 0$ for all x in (a, c) and $f'(x) > 0$ for all x in (c, b), then $f(c)$ is a local minimum value of f.
3) If $f'(x)$ has the same sign on both sides of c, then $f(c)$ is not a local extreme value of f.

Local Extrema

Proof of 1).

□

Since $f'(x) > 0$ for all x in (a, c), f is increasing on $(a, c]$ by the Monotonicity theorem.

And again, since $f'(x) < 0$ for all x in (a, c), f is decreasing on $[c, b)$ by the Monotonicity theorem.

Thus, $f(x) < f(c)$ for all x in (a, b), except of course at $x = c$.

We conclude that $f(c)$ is a local maximum.

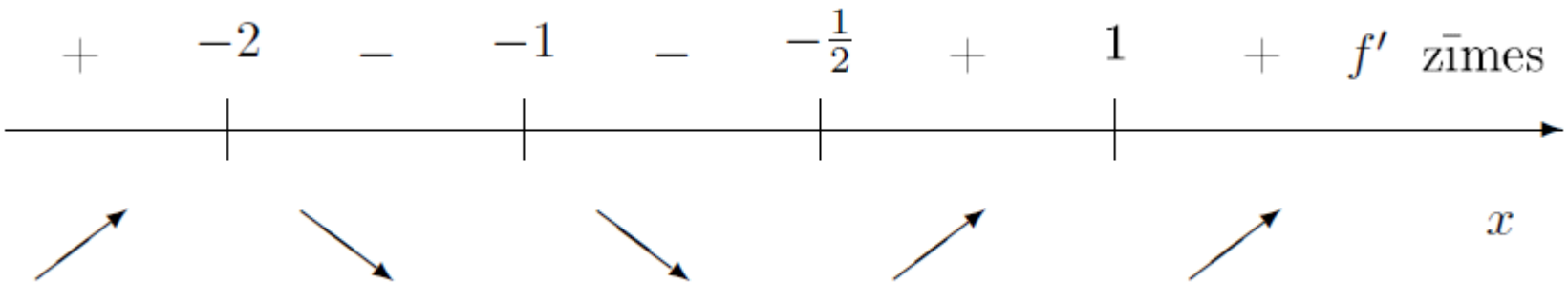
■

Local Extrema

Exercise

$$f(x) = \frac{4x+5}{1-x^2} \qquad f'(x) = \frac{4(1-x^2)+2x(4x+5)}{(1-x^2)^2} = \frac{2(2x^2+5x+2)}{(1-x^2)^2}$$

Critical points are -2, -1/2, -1 and 1.



At -1 and 1 there is no extrema values, because the derivative values do not change signs. Also these values do not belong to Dom(f(x)).

According the change of signs at the point -2 the function has local maximum value 1 and at the -1/2 the function has local minimum value 4.
We see that these are local extrema values, because the maximum value is smaller then minimum value.

Local Extrema

Exercise

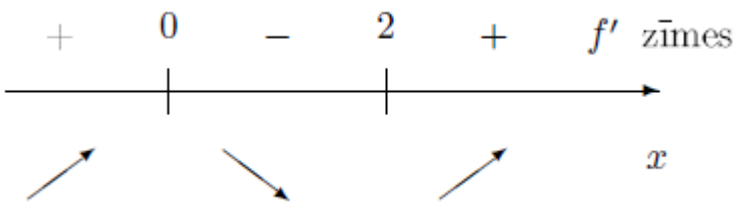
$$f(x) = \frac{x^3 + 4}{x^2}$$

Please find the extrema values!

$$y' = \frac{x^3 - 8}{x^3}$$

Critical points are 0 and 2.

Hav to detect signd for first derivative.



At 0 the function is not defined, thus it cannot have extrema value there, even if the derivative changes the sign.

The point (2, 3) in accordance with the change of signs is a local minimum point.

It is local, because, for example) f(-2) = -1 < 3.t

Concavity

Exercises

Please find the extreme values! Please detect if any of them is the global extreme value!

$$a)\ y = \frac{lnx}{x}$$

$$b)\ y = \frac{x^n}{1 + x^2}, n = 1, 2, \dots,$$

HOMEWORK 8

15. Where is $y = \frac{x}{(x-2)^2}$ concave up and where it is concave down? Please, detect th points of inflection.

Answer:
point of inflection (-4; -1/9)
oncave down on (-4;2) and (2; +∞)
concave up on (- ∞;-4)

16. The the number a in the sum with the square of the number b gives 48. Please, find such a and b, so that their multiplication is greatest.

Answer: a=32, b=4

Reading: D. Varberg, E. J. Purcell, S. E. Rigdon *Calculus*. 9th Edition, Prentice-Hall, Inc.m, 2014, p. 155– 166.
Chapter 3. Sections 3.2., 3.3.